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Introduction

DSQ is the native, binary format used by the Torque engine to store animation data. This document serves as a basic specification for the DSQ file format. Very little context is given to the sequence data fields read from the DSQ file so it is recommended that this resource be read along with the Torque source code. The DSQ file format is shown below. Note that all values are stored as little-endian, so on non-Intel platforms you will have to handle perform endian conversion for 16 and 32-bit values.

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DSQ Format

Name	# bytes	Data Type	Description
DSQ version number	2	S16	DSQ version of this file. Note that the current DSQ version (for files saved by T3D) is 26. Most DTS exporters and the ShowToolPro tool only read/write version 24. There is actually no difference between v26 and v24, but files generated by T3D will always use the latest DTS version number.
Exporter version number	2	S16	For tracking version of the exporter that generated this file. Can usually be ignored.
numNodes	4	S32	Number of nodes in the shape
nodeNames	numNodes * ?	S32 and char[]	Node names stored as a 4-byte length followed by N characters (terminating NULL not included in length or DSQ file)
dummy	4	S32	Dummy value
numObjects	4	S32	Number of objects in the shape

numNodeRotations	4	S32	Number of node rotation keyframes
nodeRotations	numNodeRotations * 8	Quat16 { S16 x, S16 y, S16 z, S16 w }	Array of quaternions for node rotation keyframes (all sequences)
numNodeTranslations	4	S32	Number of node translation keyframes
nodeTranslations	numNodeTranslations * 12	Point3F { F32 x, F32 y, F32 z }	Array of points for node translation keyframes (all sequences)
numNodeUniformScales	4	S32	Number of node uniform scale keyframes
nodeUniformScales	numNodeUniformScales * 4	F32	Array of floats for node uniform scale keyframes (all sequences)
numNodeAlignedScales	4	S32	Number of node aligned scale keyframes
nodeAlignedScales	numNodeAlignedScales * 12	Point3F { F32 x, F32 y, F32 z }	Array of points for node aligned scale keyframes (all sequences)
numNodeArbScales	4	S32	Number of node arbitrary scale keyframes
nodeArbScaleRots	numNodeArbScales * 8	Quat16 { S16 x, S16 y, S16 z, S16 w }	Array of quaternions for node arbitrary scale rotation keyframes (all sequences)
nodeArbScaleFactors	numNodeArbScales * 12	Point3F { F32 x, F32 y, F32 z }	Array of points for node arbitrary scale factor keyframes (all sequences)
numGroundFrames	4	S32	Number of ground transform keyframes
groundTranslations	4	Point3F { F32 x, F32 y, F32 z }	Array of points for ground transform keyframes (all sequences)
groundRotations	2	Quat16 { S16 x, S16 y, S16 z, S16 w }	Array of quaternions for ground transform keyframes (all sequences)
dummy	4	S32	Dummy value
numSequences	4	S32	Number of sequences in the shape
sequences	numSequences * ?	Sequence	Array of Sequences - size of each sequence depends on which nodes are affected etc.
numTriggers	4	S32	Number of triggers (all sequences)
triggers	4	Trigger { U32 state, F32 pos }	Array of sequence triggers (all sequences)

Sequences

Sequences are stored in the DSQ file as shown below. Note that this is almost identical to how sequences are stored in [DTS files](#) except for the name:

Name	# bytes	Data Type	Description
name	?	S32 and char[]	The name of this sequence stored as a 4-byte length followed by N characters (terminating NULL not included in length or DSQ file)
flags	4	U32	Sequence flags
numKeyframes	4	S32	Number of keyframes in this sequence
duration	4	F32	Duration of the sequence (in seconds)
priority	4	S32	Sequence priority
firstGroundFrame	4	S32	First ground transform keyframe in this sequence (index into the groundTranslations and groundRotation arrays)
numGroundFrames	4	S32	Number of ground transform keyframes in this sequence
baseRotation	4	S32	First node rotation keyframe in this sequence (index into the nodeRotations array)
baseTranslation	4	S32	First node translation keyframe in this sequence (index into the nodeTranslations array)
baseScale	4	S32	First node scale keyframe in this sequence (index into the nodeXXXScales arrays)
baseObjectState	4	S32	First object state keyframe in this sequence (index into the objectStates array)
baseDecalState	4	S32	First decal state keyframe in this sequence (index into the decalStates array). Note that DTS decals are deprecated, and this value should be 0.
firstTrigger	4	S32	First trigger in this sequence (index into the triggers array)
numTriggers	4	S32	Number of triggers in this sequence
toolBegin	4	F32	Value representing the start of this sequence in the exporting tool's timeline (can usually be ignored)
rotationMatters	?	BitSet	BitSet indicating which node rotations are animated by this sequence.
translationMatters	?	BitSet	BitSet indicating which node translations are animated by this sequence.
scaleMatters	?	BitSet	BitSet indicating which node scales are animated by this sequence.
decalMatters	?	BitSet	BitSet indicating which decal states are animated by this sequence. Note that DTS decals are deprecated.
iflMatters	?	BitSet	BitSet indicating which IFL materials are animated by this sequence.
visMatters	?	BitSet	BitSet indicating which object's visibility is animated by this sequence.
frameMatters	?	BitSet	BitSet indicating which mesh's verts are animated by this sequence.
matFrameMatters	?	BitSet	BitSet indicating which mesh's UV coords are animated by this sequence.

BitSets

BitSets are stored in the DSQ file as:

Name	# bytes	Data Type	Description
dummy	4	S32	Dummy value
numWords	4	S32	Number of 32-bit values in the BitSet
bits	4 * numWords	U32	Array of 32-bit values representing the bits in the BitSet

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